COVID-19 DATA ANALYTICS DASHBOARD

Using AWS Cloud Services and Microsoft Power BI

PROJECT REPORT

A Cloud-Based Serverless Analytics Solution for Global
COVID-19 Data using Amazon Web Services and Power BI

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1. Project Overview

The COVID-19 Data Analytics Dashboard is an end-to-end cloud-based analytics project developed using Amazon Web Services (AWS) and Microsoft Power BI. The project analyzes global COVID-19 data and generates meaningful insights through SQL queries and interactive visualizations.

The project follows a serverless architecture in which data is stored in Amazon S3, metadata is managed through AWS Glue, data is queried using Amazon Athena, and insights are visualized using Power BI.

1.1 Project Objective

The main objective of this project is to build a scalable and cost-effective cloud-based analytics solution for analyzing COVID-19 data. The project demonstrates how AWS serverless services can be integrated with Power BI to perform data analysis and create interactive dashboards.

2. Dataset Information

Dataset Name	countries-aggregated.csv
Data Period	January 22, 2020 to April 16, 2022
Columns	Date, Country, Confirmed Cases, Recovered Cases, Death Cases

The dataset contains daily country-level records of confirmed, recovered, and death cases for COVID-19, spanning over two years of the pandemic.

3. Technologies Used

3.1 AWS Services

Amazon S3

Used for storing the COVID-19 dataset and Athena query results.

AWS Glue Database

Used to create a logical container for metadata.

AWS Glue Crawler

Automatically scans the dataset and creates metadata.

AWS Glue Data Catalog

Stores schema information and table definitions.

Amazon Athena

Used to execute SQL queries directly on the data stored in Amazon S3.

3.2 Business Intelligence Tool

Microsoft Power BI

Used to create dashboards and visualize the results obtained from Amazon Athena.

4. System Architecture

The diagram below illustrates the end-to-end data flow of the project, from raw dataset ingestion to final dashboard visualization.

```
COVID-19 Dataset (CSV)
     |
     v
Amazon S3
     |
     v
AWS Glue Database
     |
     v
AWS Glue Crawler
     |
     v
AWS Glue Data Catalog
     |
     v
Amazon Athena
     |
     v
Query Results
     |
     v
Microsoft Power BI Dashboard
```

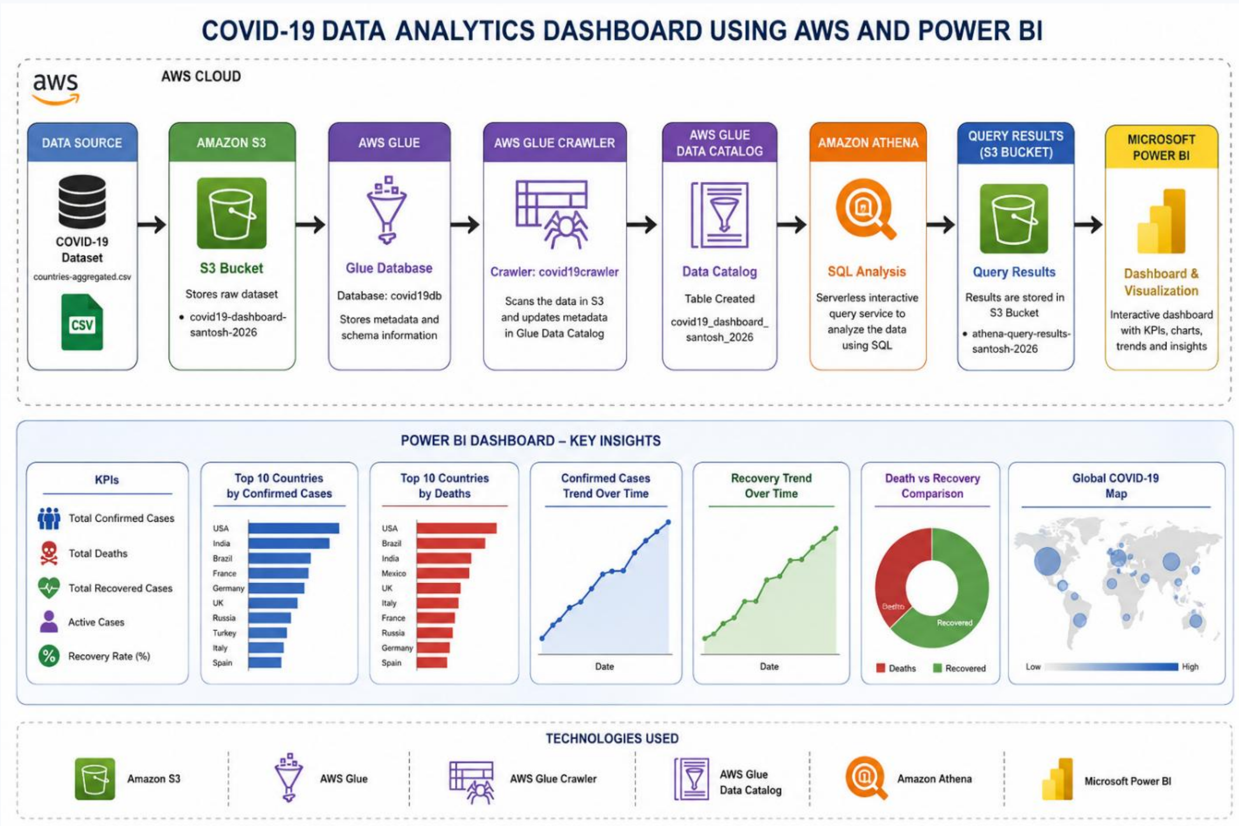

Figure 4.1: End-to-end system architecture

5. Implementation Steps

Step 1: Create S3 Bucket

Bucket Name: `covid19-dashboard-santosh-2026`

Purpose:

- Store the COVID-19 dataset.

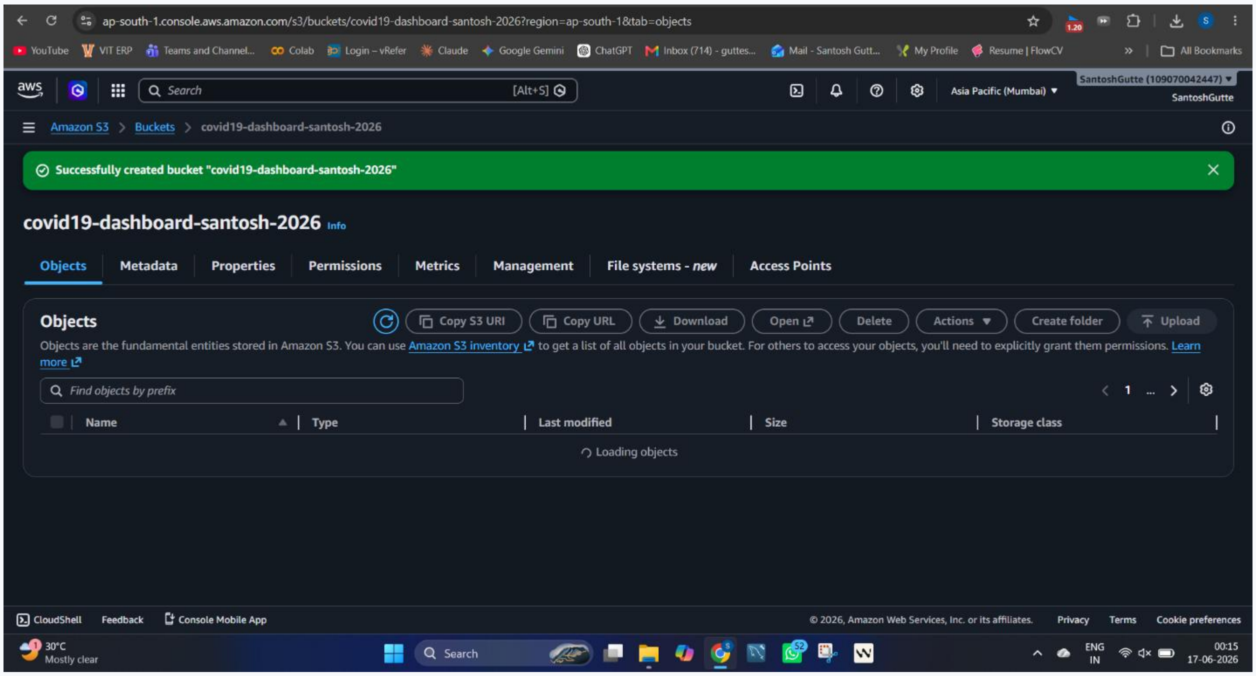

Figure: Create S3 Bucket

Step 2: Upload Dataset

File Uploaded: `countries-aggregated.csv`

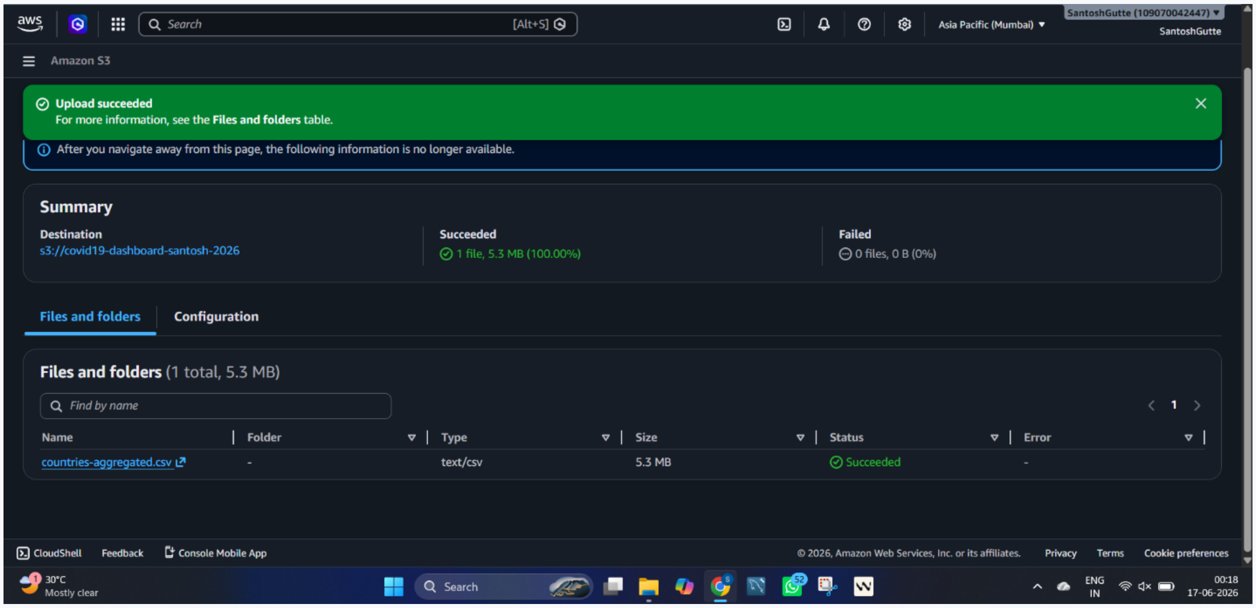

Figure: Upload Dataset

Step 3: Create Glue Database

Database Name: `covid19db`

Purpose:

- Store metadata generated by the crawler.

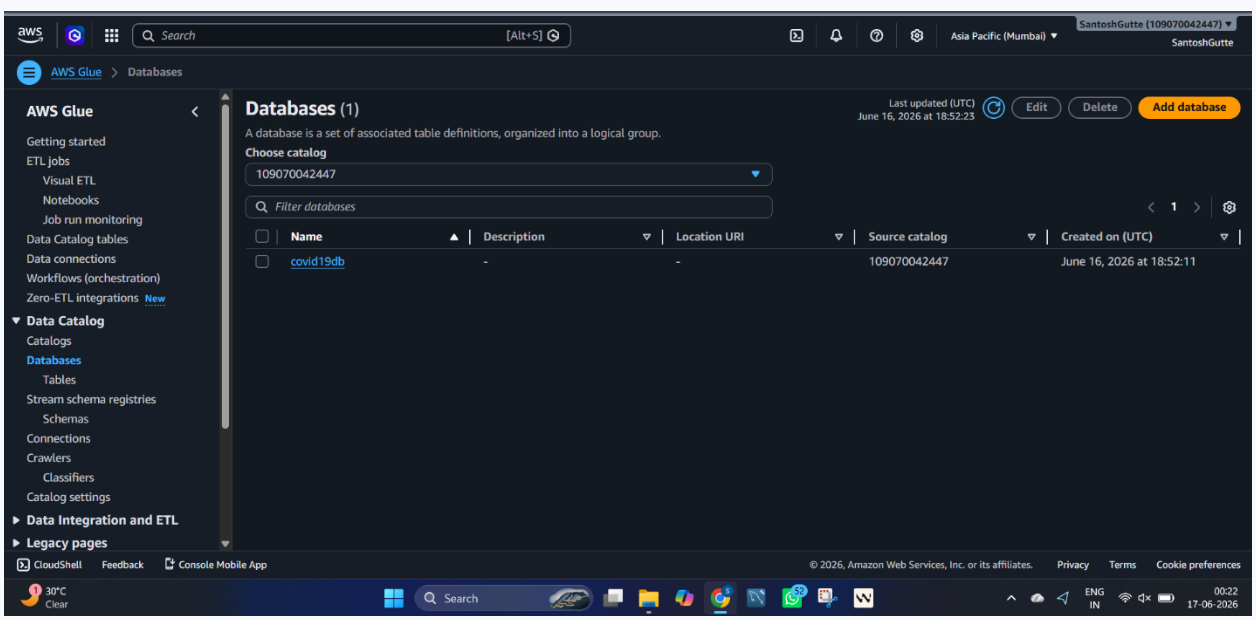

Figure: Create Glue Database

Step 4: Create AWS Glue Crawler

Crawler Name: `covid19crawler`

Purpose:

- Scan the dataset stored in S3.
- Automatically detect schema.
- Create metadata.

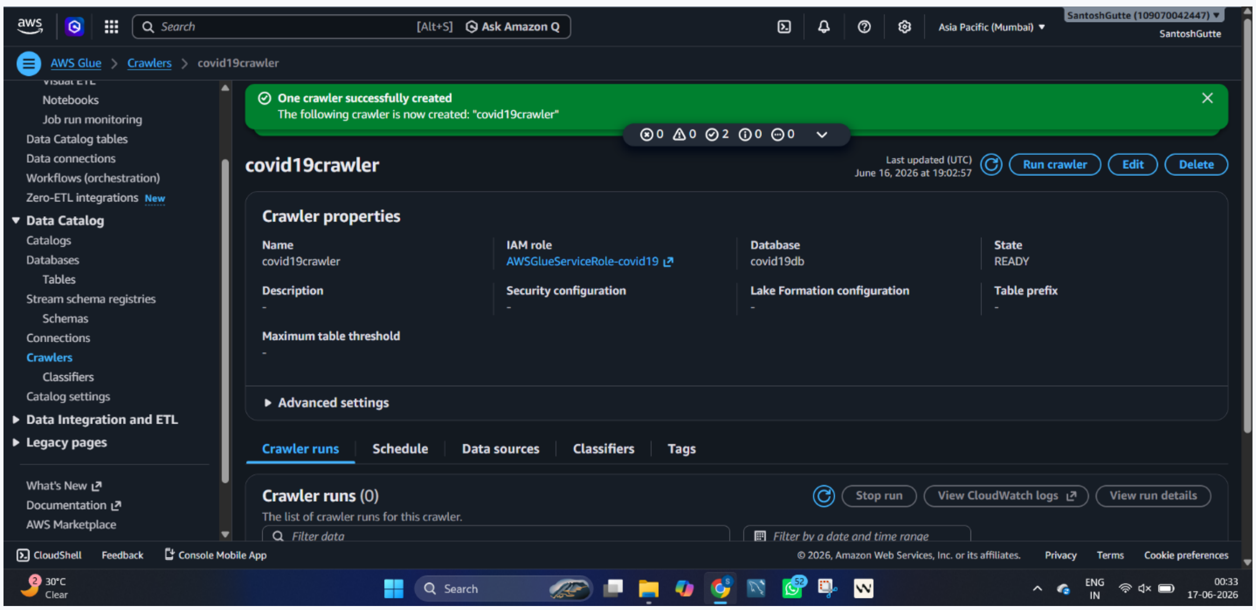

Figure: Create AWS Glue Crawler

Step 5: Configure Data Source

S3 Path: `s3://covid19-dashboard-santosh-2026`

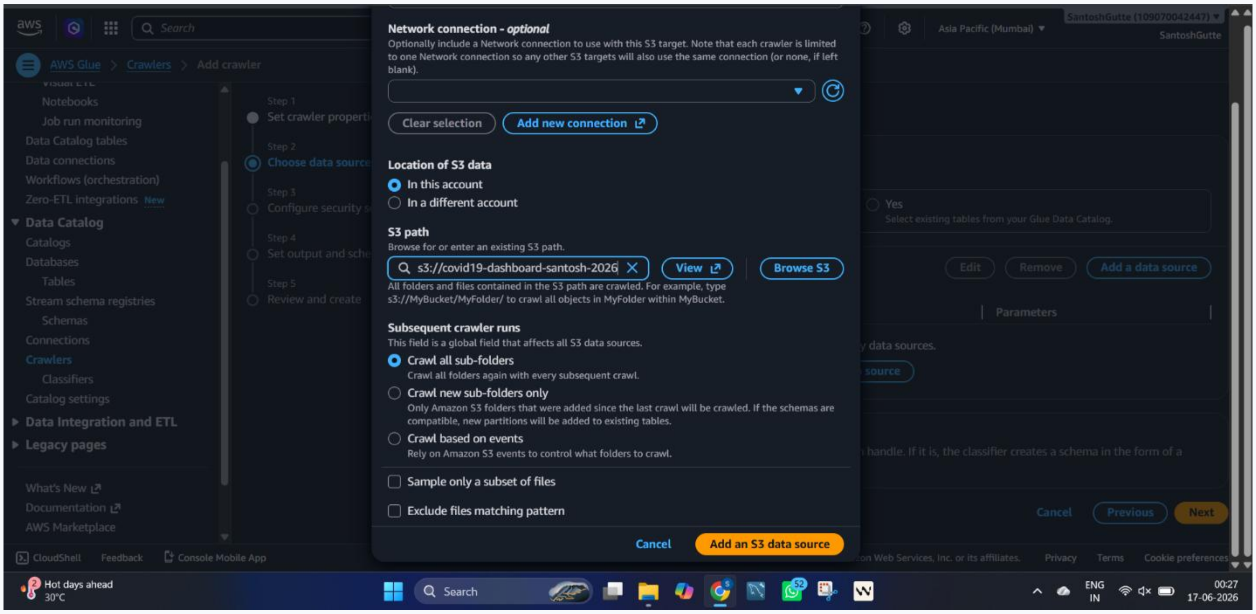

Figure: Configure Data Source

Step 6: Configure IAM Role

IAM Role: `AWSGlueServiceRole-covid19`

Purpose:

- Grant AWS Glue permission to access S3 resources.

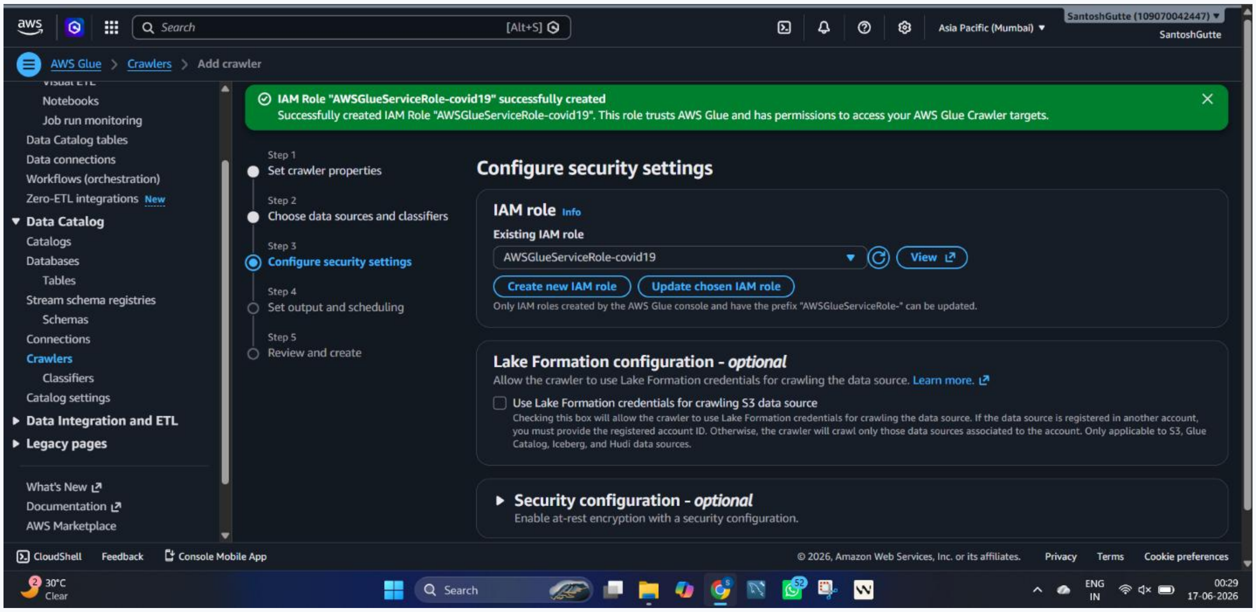

Figure: Configure IAM Role

Step 7: Create Data Catalog Table

Table Name: `covid19_dashboard_santosh_2026`

Purpose:

- Make the dataset queryable through Athena.

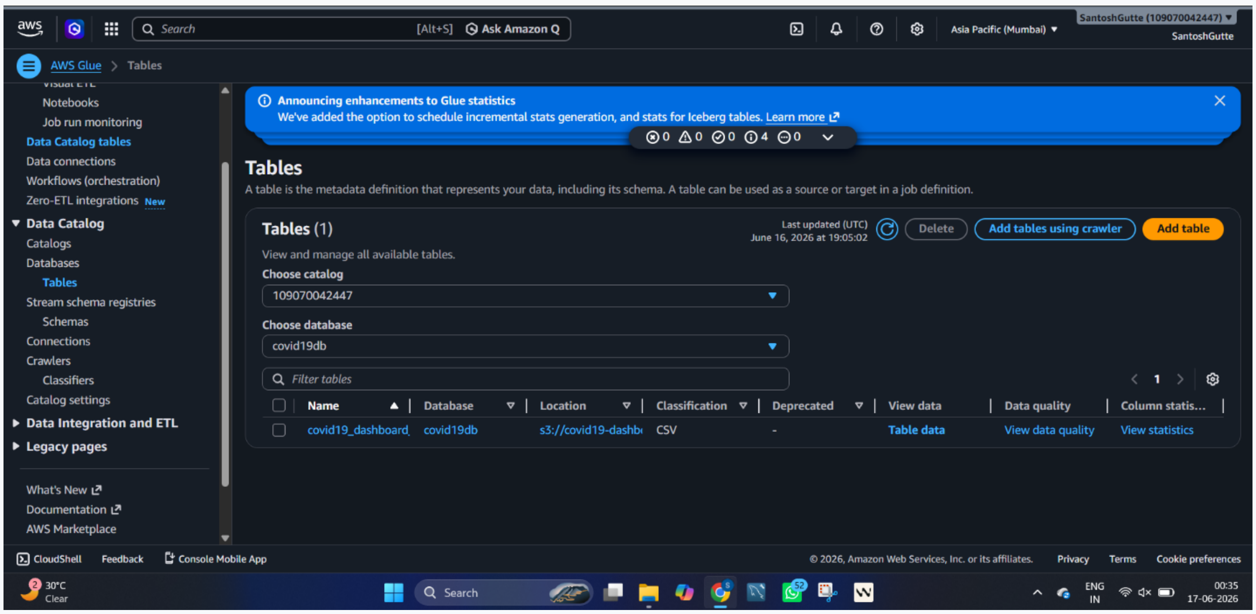

Figure: Create Data Catalog Table

Step 8: Create Query Result Bucket

Bucket Name: `athena-query-results-santosh-2026`

Purpose:

- Store query outputs generated by Athena.

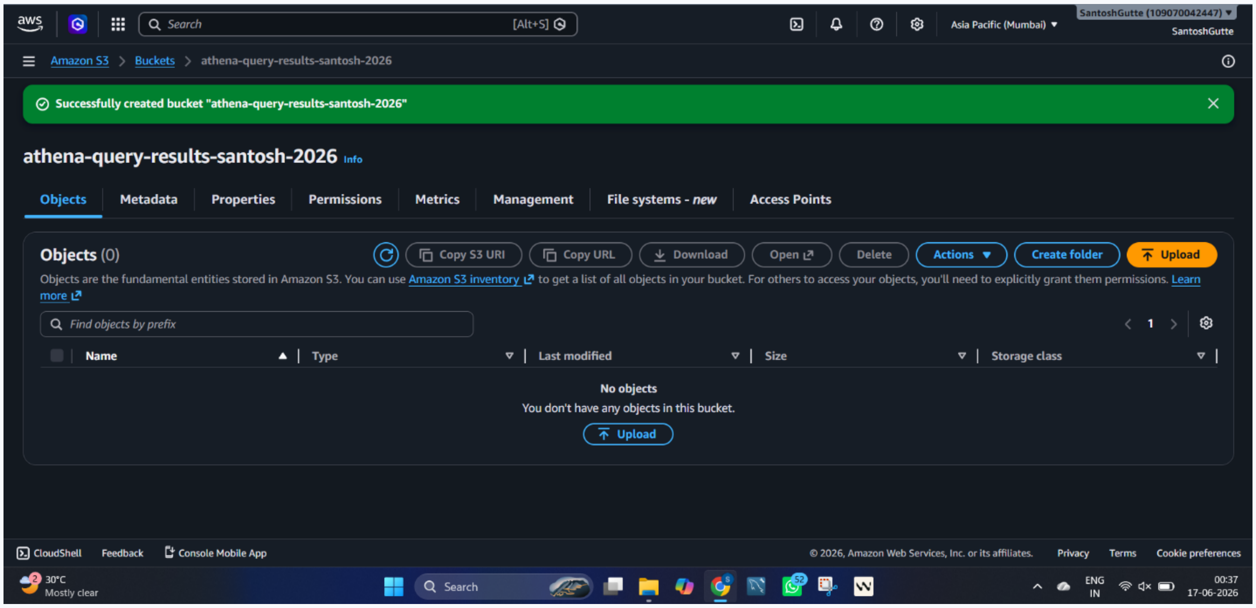

Figure: Create Query Result Bucket

Step 9: Configure Amazon Athena

Query Result Location: `s3://athena-query-results-santosh-2026`

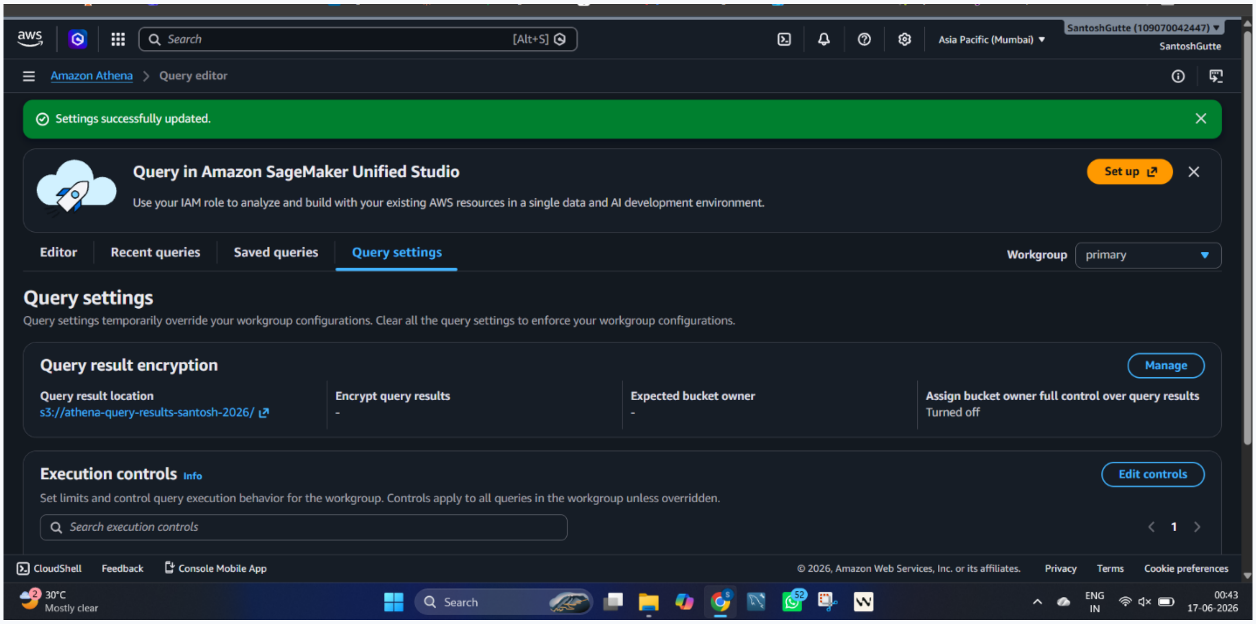

Figure: Configure Amazon Athena

6. SQL Queries Performed
The following SQL queries were executed in Amazon Athena against the cataloged dataset to extract insights.

6.1 Display First 10 Records

```
SELECT *
FROM covid19_dashboard_santosh_2026
LIMIT 10;
```

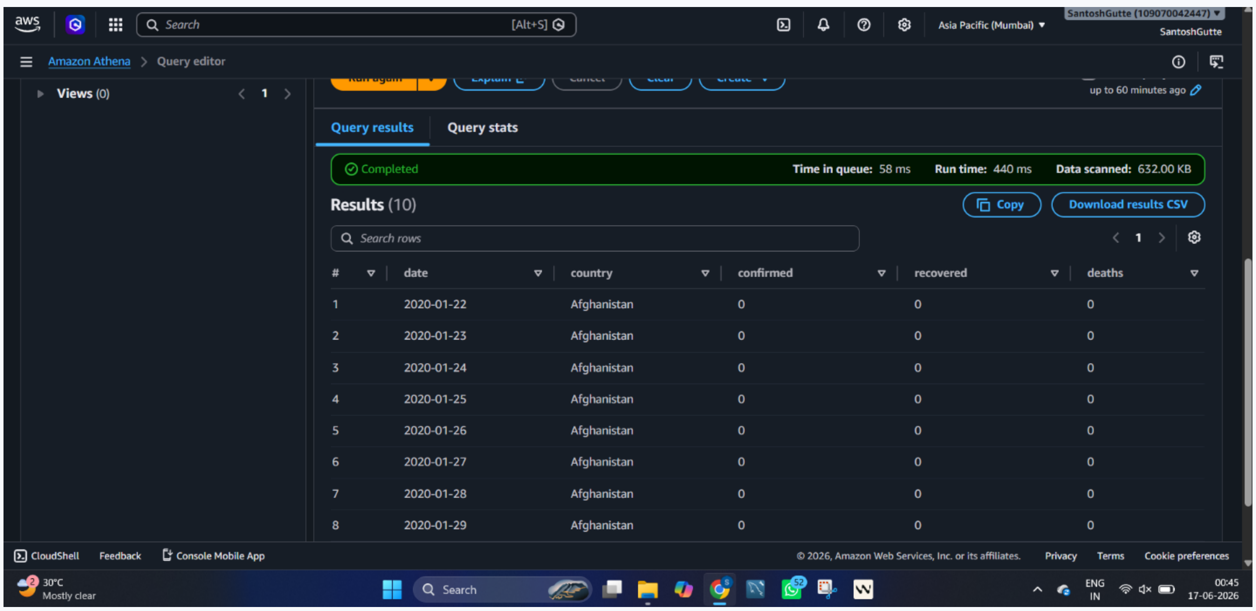

Figure 6.1: Display First 10 Records — Athena query result

6.2 Top 10 Countries by Confirmed Cases

```
SELECT country,
MAX(confirmed) AS total_cases
FROM covid19_dashboard_santosh_2026
GROUP BY country
ORDER BY total_cases DESC
LIMIT 10;
```

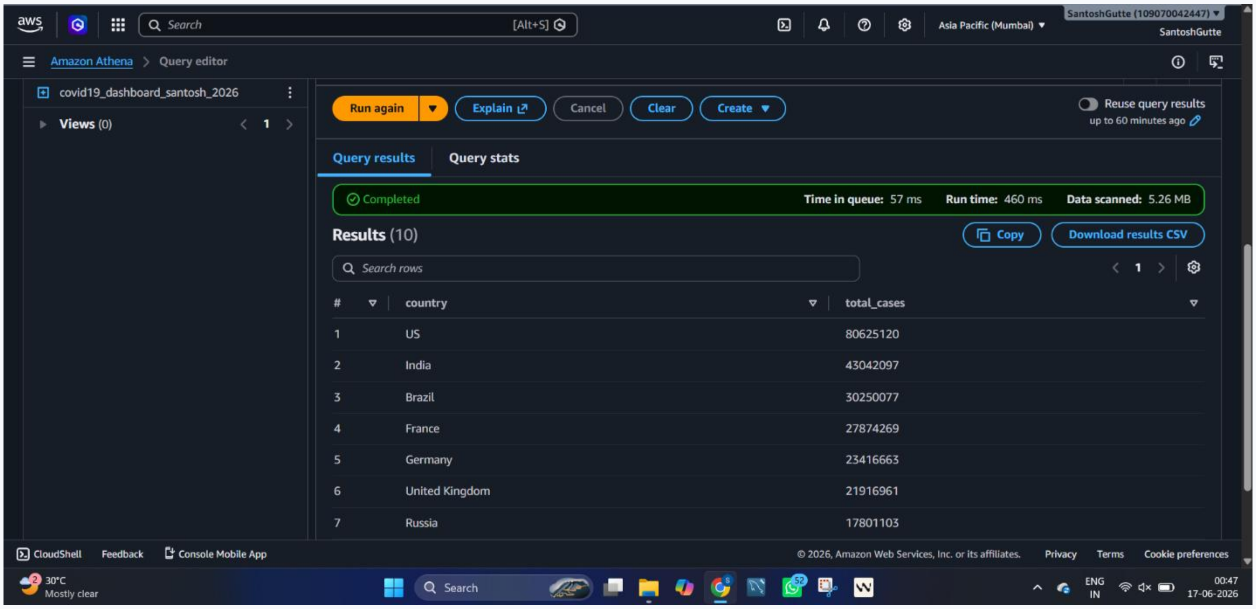

Figure 6.2: Top 10 Countries by Confirmed Cases — Athena query result

6.3 Top 10 Countries by Death Cases

```
SELECT country,
MAX(deaths) AS total_deaths
FROM covid19_dashboard_santosh_2026
GROUP BY country
ORDER BY total_deaths DESC
LIMIT 10;
```

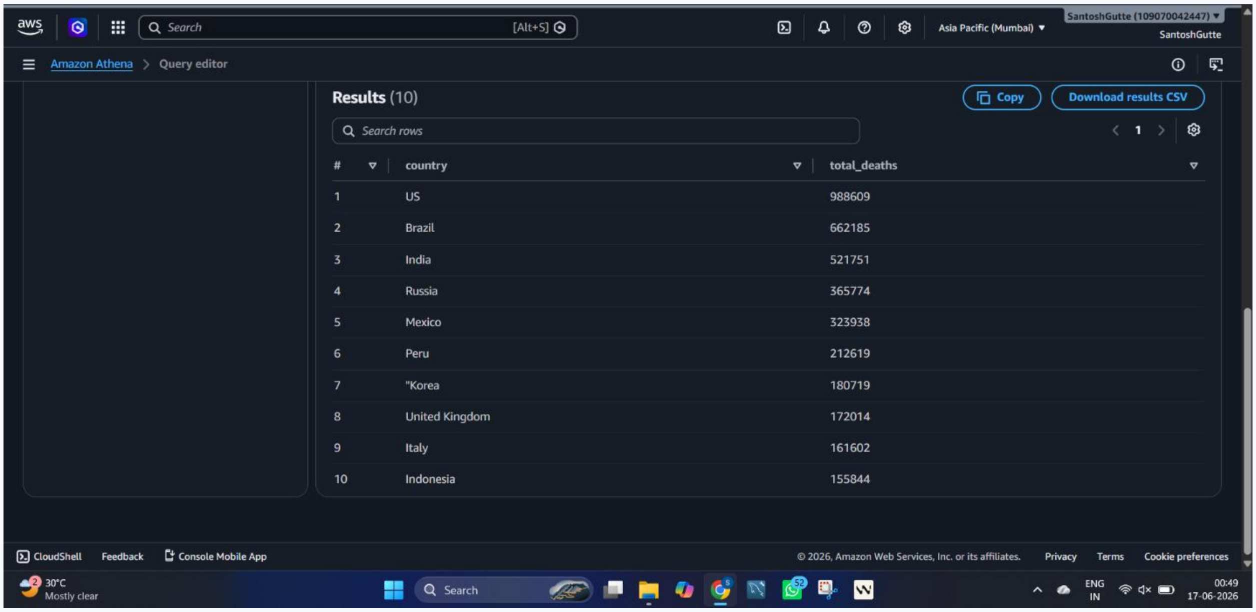

Figure 6.3: Top 10 Countries by Death Cases — Athena query result

6.4 Top 10 Countries by Recovery Cases

```
SELECT country,
MAX(recovered) AS total_recovered
FROM covid19_dashboard_santosh_2026
GROUP BY country
ORDER BY total_recovered DESC
LIMIT 10;
```

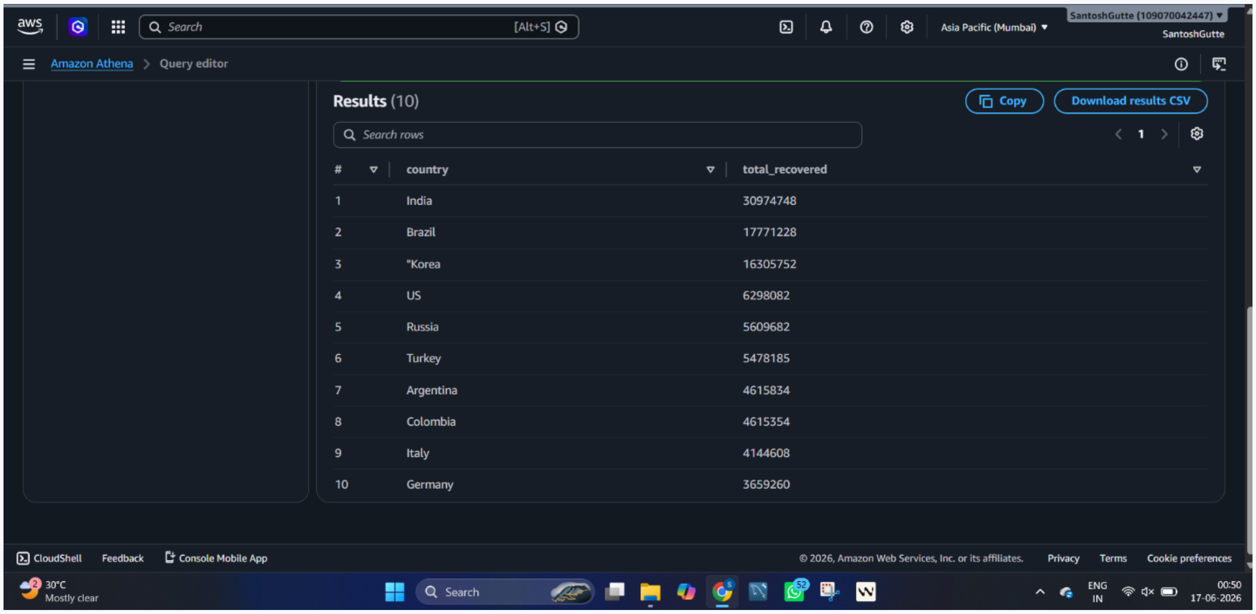

Figure 6.4: Top 10 Countries by Recovery Cases — Athena query result

6.5 Global Statistics

```
SELECT
SUM(confirmed) AS total_confirmed,
SUM(recovered) AS total_recovered,
SUM(deaths) AS total_deaths
FROM covid19_dashboard_santosh_2026;
```

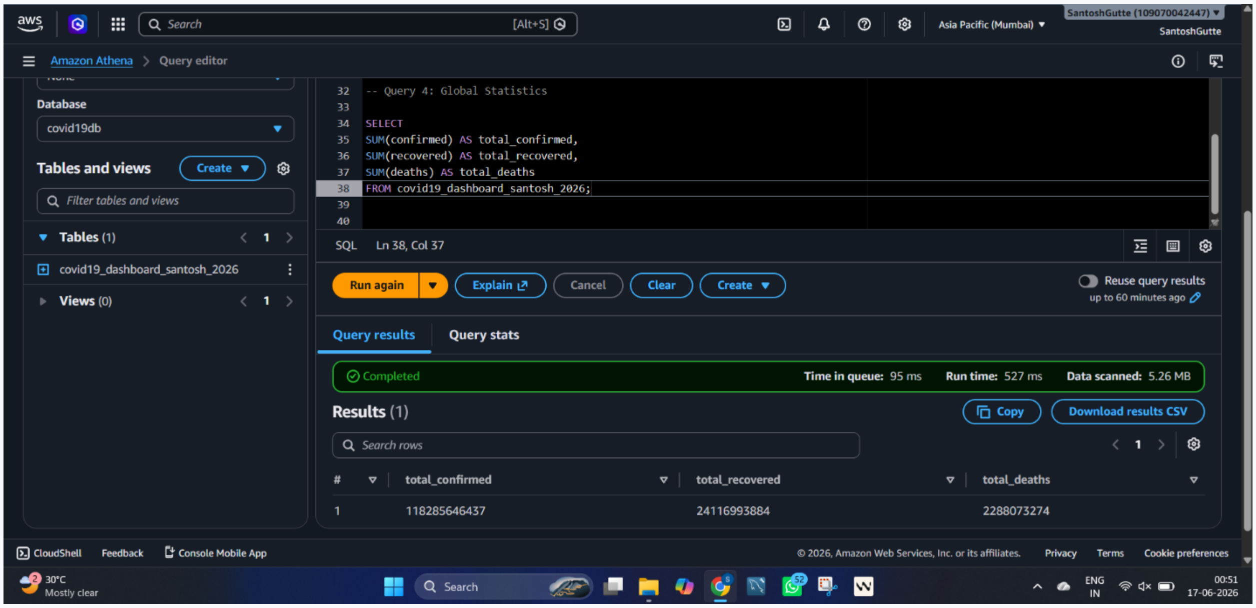

Figure 6.5: Global Statistics — Athena query result

6.6 Daily Trend Analysis

```
SELECT date,
SUM(confirmed) AS total_cases
FROM covid19_dashboard_santosh_2026
GROUP BY date
ORDER BY date;
```

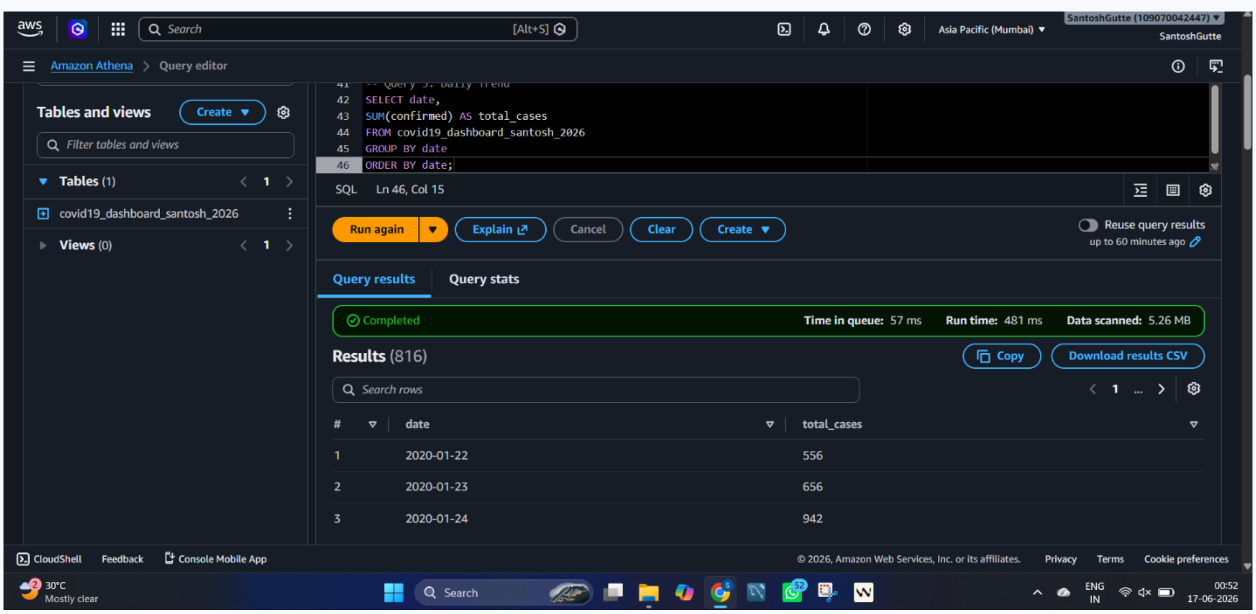

Figure 6.6: Daily Trend Analysis — Athena query result

7. Power BI Dashboard
The query results obtained from Amazon Athena were imported into Power BI to build the following visualizations.

7.1 KPI Cards

- Total Confirmed Cases
- Total Death Cases
- Total Recovery Cases
- Active Cases
- Recovery Rate

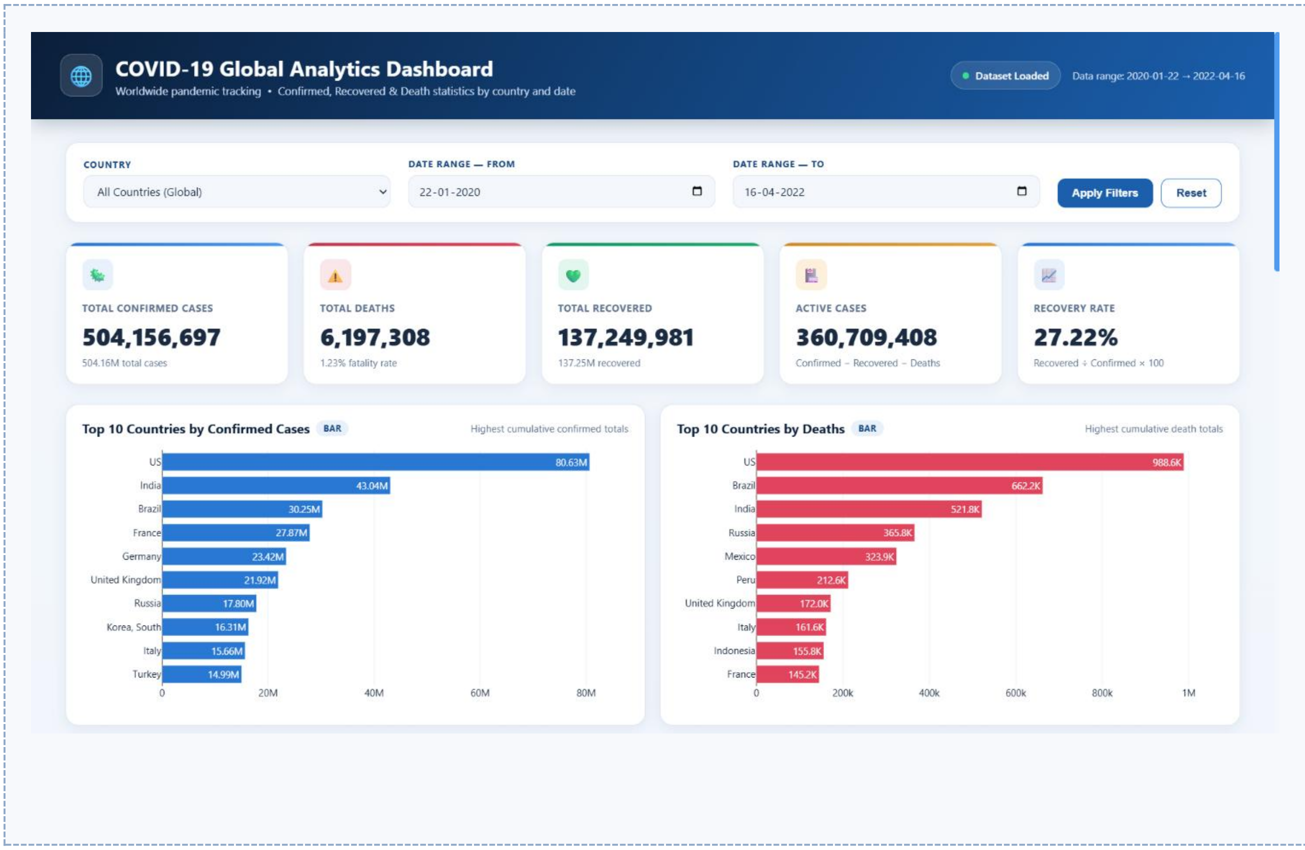

Figure 7.1: KPI cards

7.2 Charts Used

- Bar Chart — Top 10 Countries by Confirmed Cases
- Bar Chart — Top 10 Countries by Death Cases
- Bar Chart — Top 10 Countries by Recovery Cases
- Line Chart — Daily Confirmed Cases Trend
- Area Chart — Recovery Trend

- Donut Chart — Deaths vs Recoveries
- Table Visualization — Country-wise Statistics

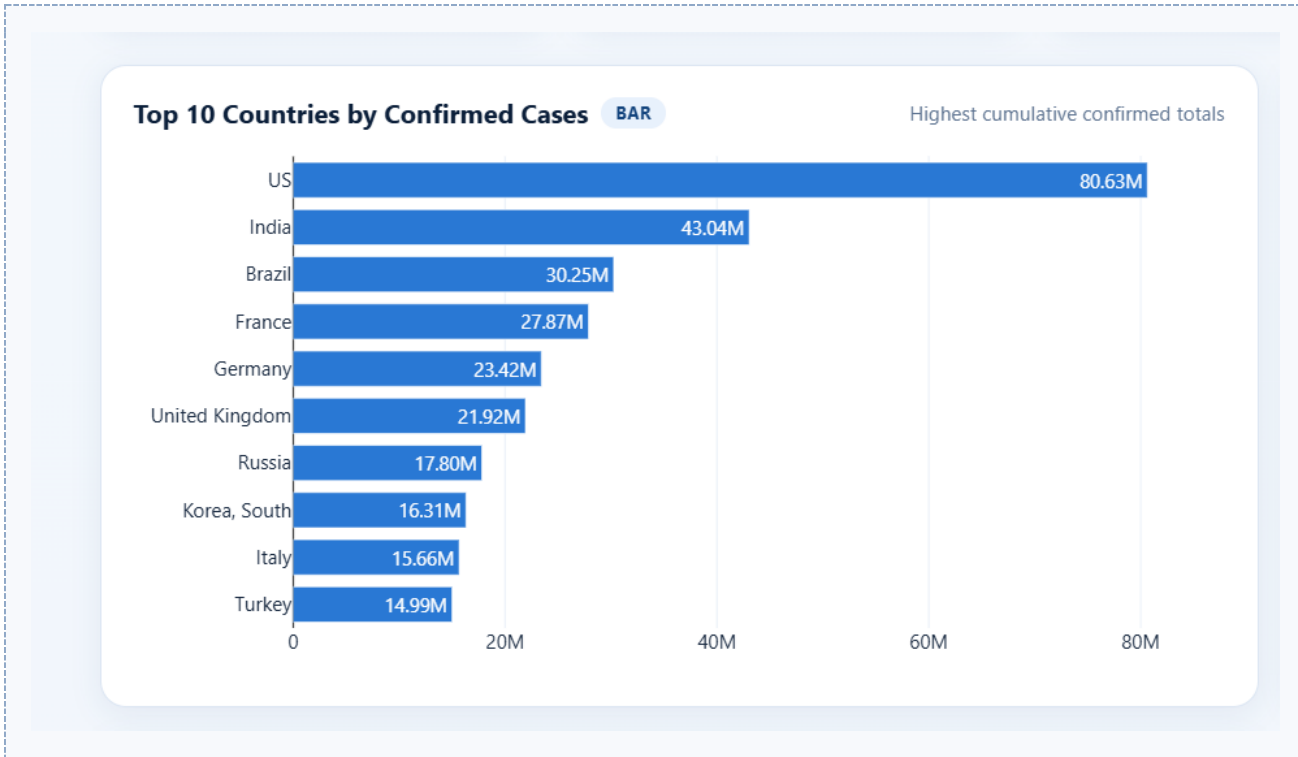

Figure 7.2: Top 10 countries — bar chart visualizations

Figure 7.3: Daily confirmed cases trend

Figure 7.4: Recovery trend

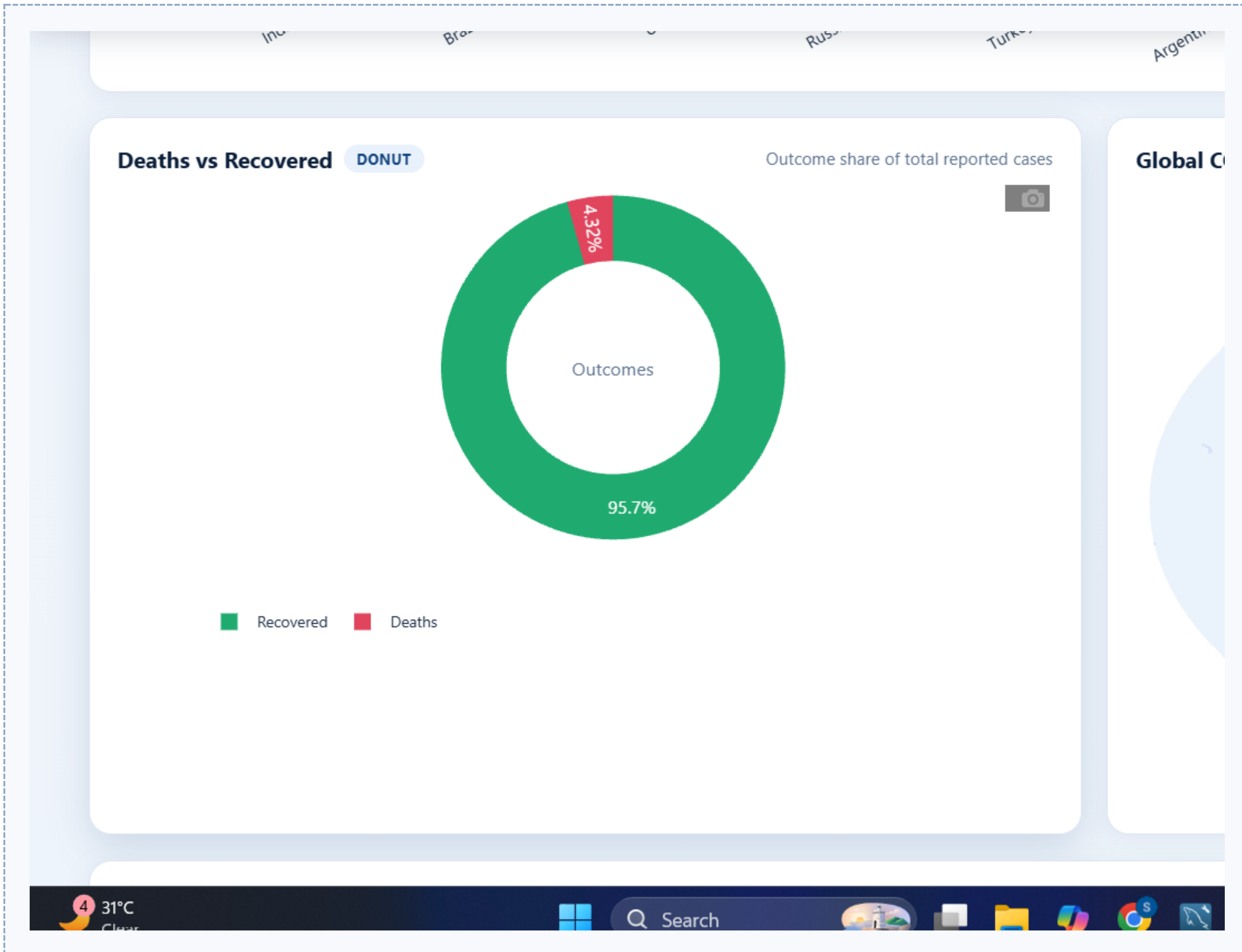

Figure 7.5: Deaths vs recoveries

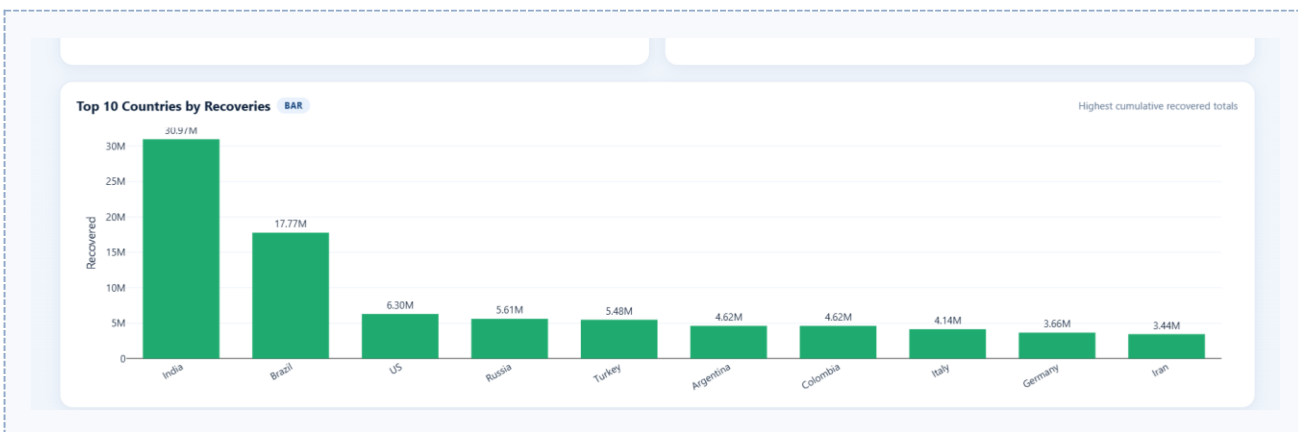

Figure 7.6: Country-wise statistics table

7.3 Filters

- Country Filter
- Date Filter

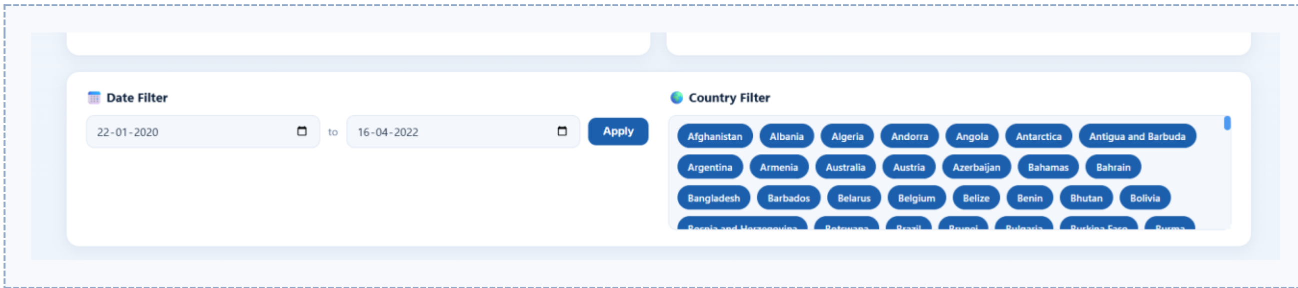

Figure 7.7: Complete Power BI dashboard

8. Key Insights

- United States had the highest confirmed cases.
- India recorded the highest recoveries.
- Daily trends showed a rapid increase in cases during multiple waves.
- Recovery rates improved significantly over time.
- Global statistics provided an overview of the pandemic's impact.

9. Advantages

- Serverless architecture — no infrastructure management.
- Highly scalable.
- Cost-effective.
- Fast query execution.
- Easy integration with Power BI.

10. Applications

- Healthcare analytics.
- Pandemic monitoring.
- Research and statistical analysis.
- Government reporting.
- Decision support systems.

11. Future Scope

- Real-time data integration.
- Machine learning-based prediction models.
- Automated report generation.
- Interactive cloud dashboards.
- Big data analytics integration.

12. Conclusion

The COVID-19 Data Analytics Dashboard successfully demonstrates how AWS services and Power BI can be integrated to build an end-to-end cloud analytics solution. The project showcases serverless data storage, metadata management, SQL-based analysis, and business intelligence visualization, making it an excellent example of practical cloud computing and data analytics.